

## Problem-Solution Fit

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| <b>Date</b>          | 15 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-5638                 |
| <b>Project Name</b>  | Credit card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

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| <b>1. CUSTOMER SEGMENT(S) [CS]</b> <ul style="list-style-type: none"> <li>• Individuals applying for a credit card</li> <li>• Banks and financial institutions</li> <li>• Loan/Credit officers</li> </ul>                                      | <b>6. CUSTOMER LIMITATIONS</b><br>EG. BUDGET, DEVICES [CL] <ul style="list-style-type: none"> <li>• Limited financial knowledge</li> <li>• Lack of credit history</li> <li>• Time constraints</li> <li>• Internet/device availability</li> </ul>                            | <b>5. AVAILABLE SOLUTIONS</b><br>PROS & CONS [AS] <b>Manual Verification</b> <ul style="list-style-type: none"> <li>✓ Human judgment</li> <li>✗ Slow and error-prone</li> </ul> <b>Traditional Rule-Based Systems</b> <ul style="list-style-type: none"> <li>✓ Simple to implement</li> <li>✗ Less accurate and inflexible</li> </ul> |
| <b>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</b> <ul style="list-style-type: none"> <li>• Manual approval process is slow (High)</li> <li>• Inconsistent decision-making (Medium)</li> <li>• Long waiting time for applicants (High)</li> </ul> | <b>9. PROBLEM ROOT / CAUSE [RC]</b> <ul style="list-style-type: none"> <li>• Large number of applications • Time-consuming manual verification</li> <li>• Human bias and inconsistency</li> <li>• Difficulty analyzing multiple applicant factors simultaneously</li> </ul> | <b>7. BEHAVIOR + ITS INTENSITY [BE]</b> <ul style="list-style-type: none"> <li>• Frequently checks eligibility before applying (High)</li> <li>• Wants quick decisions (High)</li> <li>• Compares different banks (Medium)</li> </ul>                                                                                                 |

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| <p><b>3. TRIGGERS TO ACT [TR]</b></p> <ul style="list-style-type: none"> <li>Customer applies for a credit card</li> <li>Need for instant eligibility</li> </ul>              | <p><b>10. YOUR SOLUTION [SL]</b></p> <p>Develop a <b>Machine Learning-based Credit Card Approval Prediction</b></p>                                                                                                                            | <p><b>8. CHANNELS OF BEHAVIOR [CH]</b></p> <p>ONLINE<br/>Bank websites, Mobile apps, Credit card portals</p> |
| <p>check</p> <ul style="list-style-type: none"> <li>Bank wants to automate approvals</li> </ul>                                                                               | <p><b>System</b> using <b>Random Forest</b> and <b>Flask</b> to analyze applicant details and predict approval/rejection instantly. The system provides faster, consistent, and data-driven decisions through an easy-touse web interface.</p> |                                                                                                              |
| <p><b>4. EMOTIONS BEFORE / AFTER [EM]</b></p> <p><b>Before:</b> Anxious, uncertain, hopeful</p> <p><b>After:</b> Happy if approved; disappointed but informed if rejected</p> |                                                                                                                                                                                                                                                | <p>OFFLINE<br/>Bank branches, Customer service representatives</p>                                           |